WASP: Watchlist Attack Shield Protocol for Privacy-Preserving Information-centric Networking (ICN)

Kumari Nidhi Lal, Krishna Malani
Department of Computer Science & Engineering,
Visvesvaraya National Institute of Technology, Nagpur
Maharashtra, 440010, India
nidhilal@cse.vnit.ac.in, krm_icn@outlook.com

Abstract: Information-Centric Networking (ICN) has emerged as a promising future Internet architecture by enabling name-based content retrieval, in-network caching, and efficient dissemination of content. However, this new paradigm exposes ICN subscribers to a watchlist attack, in which adversaries monitor the pattern of content requests and sensitive information without disrupting normal network operations. Existing solutions primarily address Interest Flooding and Content Poisoning attacks while providing limited protection against the privacy leakage caused by watchlist attacks. Therefore, motivated by this, this paper proposes *WASP* (*Watchlist Attack Shield Protocol*) that integrates intelligent traffic monitoring, lightweight dynamic cryptographic protection, and secure forwarding to mitigate watchlist attacks. WASP continuously analyzes content request behavior to estimate privacy exposure and selectively activates cryptographic shielding and forwarding adaptation according to the observed threat level. In this way, it minimizes information leakage while preserving network performance. A mathematical model is developed to characterize the adaptive shielding mechanism, followed by theoretical analyses of computational complexity and security against passive traffic observation. Extensive simulations demonstrate the effectiveness of WASP under watchlist attack scenarios. To further confirm the validity, WASP is compared with existing state-of-the-art approaches. The results demonstrated that WASP significantly reduces latency degradation and routing overhead while effectively mitigating privacy leakage arising from traffic analysis.

***Index Terms*—Information-Centric Networking (ICN), Named Data Networking (NDN), Watchlist Attack, Privacy Preservation, Secure Forwarding, Adaptive Security, Cryptographic Protection, Traffic Analysis, Network Security.**

I. Introduction

The rapid growth of content-oriented Internet applications, including video streaming, cloud services, Internet of Things (IoT), and edge computing, has exposed the scalability limitations of the traditional host-centric Internet/TCP/IP architecture. Existing IP-based communication focuses on the IP address rather than the content itself. Therefore, this architecture leads to increased latency, inefficient bandwidth utilization, and complex mobility management. Information-Centric Networking (ICN) has emerged as a promising future Internet architecture to address these challenges by providing content name-based communication and ubiquitous in-network caching [8], [12], [1].

Such properties make ICN efficient; nonetheless, they simultaneously bring significant security and privacy problems. As a consequence, Content Routers (CRs) store frequently requested a consequence, Content opportunistically; attackers can passively deduce sensitive user information. The literature review indicates that ICN is notably at risk for privacy infringements such as cache privacy attacks, timing analysis attacks, and watchlist attacks [12], [11]. Watchlist attacks weaken user privacy by exploiting the openness of content nomenclature and cache behavior, rendering them challenging to identify with existing detection and mitigation strategies.

In addition, a watchlist attack typically consists of a passive adversary maintaining a predefined list of sensitive content names and monitoring Interest packets traversing one or more CRs. By monitoring, the adversary can infer user interests, browsing habits, organizational activities, and confidential communications [3]. Since this attack is passive in nature, it produces minimal network anomalies and is difficult to capture. As ICN deployment expands toward smart cities, healthcare, industrial IoT, military communications, and edge computing environments, protecting user privacy against watchlist attacks has become an increasingly important research challenge [4].

Existing ICN security research has predominantly been devoted to mitigating Interest Flooding Attacks (IFA), Content Poisoning Attacks (CPA), cache pollution attacks, forwarding attacks, and Denial-of-Service (DoS) attacks through rate limiting, machine learning, cryptographic verification, and trust management [2], [12]. These approaches effectively contribute to active attacks that manipulate network resources by analyzing some signatures. In the era of encryption mechanisms, they can protect content payloads but do not sufficiently conceal request behavior [5]. Consequently, the literature survey illustrates that a significant research gap remains in designing lightweight and adaptive mechanisms that preserve user privacy in terms of watchlist attacks.

Motivated by this, a solution named WASP (Watchlist Attack Shield Protocol) has been developed as an adaptive privacy-preserving framework for mitigating watchlist attacks. The proposed strategy WASP continuously evaluates the privacy exposure associated with content requests and dynamically adjusts its shielding strategy according to the observed threat level [7]. The proposed framework integrates various steps like intelligent traffic monitoring, adaptive secure forwarding, and lightweight cryptographic protection to minimize information leakage. By selectively activating privacy protection based on the requirement, WASP significantly reduces computational overhead and maintains high network performance under both normal and adversarial conditions.

The proposed framework WASP is supported by a mathematical model that quantifies privacy exposure based on request behavior [6]. Extensive simulations demonstrate that WASP effectively preserves network performance while significantly improving user privacy by mitigating watchlist attacks. Experimental results show that high detection accuracy, together with improvements in various key performance indicators, are achieved by the proposed work under watchlist attack scenarios.

The contributions of this paper are summarized as follows:

- First, the security and privacy implications of watchlist attacks are investigated in ICN and analyzed their impact on content confidentiality, cache privacy, and user anonymity.
- Second, a novel strategy WASP, is proposed as a privacy-preserving shield protocol that combines intelligent traffic monitoring, adaptive secure forwarding, and lightweight crypto-

graphic protections using proofs and lemmas.
- Further, an integrated mathematical formulation for adaptive privacy-risk estimation and shielding decisions is proposed for theoretical analyses of the computational complexity of the proposed strategy.
- Extensive simulations are performed under normal and adversarial network conditions. In this simulation, multiple network performance like in terms of privacy, cryptographic metrics, latency, Packet Delivery Ratio (PDR), Cache Hit Ratio (CHR), Interest Satisfaction Rate (ISR), hop count, detection accuracy, entropy, encryption overhead, and decryption overhead, are evaluated. For further validation, results are compared with state-of-the-art approaches.

The remainder of this paper is organized as follows. Section II reviews related work on ICN security and privacy. Section III presents the system architecture and threat model. Section IV describes the proposed WASP framework and its mathematical formulation. Section V provides theoretical analysis and security proofs. Section VI presents the simulation setup and performance evaluation. Section VII discusses the experimental results, and Section VIII concludes the paper with future research directions.

II. Literature Survey

ICN has been widely investigated as a future Internet architecture that shifts communication from host-based addressing to name-based content retrieval. Xylomenos *et al.* [8] presented a comprehensive survey of ICN research and discussed the major architectural components, including naming, routing, caching, and security. Referring to visibility, the security challenges have been exposed. Ahlgren *et al.* [9] further explained the fundamental principles of ICN and highlighted its potential over current host/IP based architechture to improve scalability, mobility support, and content distribution efficiency. A survey by Yu *et al.* [10] has highlighted security and privacy-related attacks that are still important challenges for the real-time adoption of ICN.

AbdAllah *et al.* [11] surveyed major ICN attacks, including DoS, IFA, CPA, and privacy attacks. This security-related survey of ICN provides intrinsic solutions through content signatures and involvement of public-key cryptography. Tourani *et al.* present a comprehensive survey on privacy, authentication, authorization and access control mechanisms in ICN is presented in [12]. They categorized ICN privacy threats into user privacy, name privacy, signature privacy, and content privacy. In this work, the authors have shown that conventional TCP/IP security solutions are insufficient for protecting ICN subscribers.

In the other boat, caching is one of the most important ICN features, which makes it independent and different from traditional IP architecture. However, the feature of caching also creates privacy risks. Zhang *et al.* [13] surveyed caching mechanisms in ICN and discussed how cache placement/replacement strategies influence latency, bandwidth consumption, and content availability. Further studies are carried out to indicate that a bad deal of cache behavior can also leak user access patterns. Acs *et al.* [15] studied privacy-aware caching in ICN and showed that cache responses may reveal whether specific content has been previously requested. Referring to the pattern of cache responses, it can be seen that this feature is highly relevant to watchlist attacks. Furthermore, timing-based privacy attacks have also received attention in the literature survey. Mohaisen *et al.* [14] investigated timing attacks on access privacy in ICN. The corresponding study demonstrated that adversaries can infer content access frequency patterns by observing timing differences caused by caching and forwarding. This observation also states the example of watchlist attacks, where the adversary passively monitors request timing and cache responses.

To improve trustworthiness, Anisetti *et al.* Anisetti et al. [16] proposed a security certification scheme for ICN to verify network security properties continuously. However, this scheme does not directly address passive privacy leakage caused by watchlist-based monitoring. Overall, the literature shows that existing state-of-the-art studies have significantly contributed to caching, security, and privacy protection. However, most existing mechanisms focus on general privacy, cache privacy, timing attacks, or active attacks, but the defense strategy against watchlist attacks remains insufficiently explored. Motivated by this gap, this paper proposes WASP for privacy-preserving ICN. Unlike static privacy-preserving mechanisms in state-of-the art literature, WASP dynamically estimates privacy exposure and adaptively activates shielding, cryptographic protection, and secure forwarding to reduce information leakage without compromising the key performanace indicators.

III. Proposed WASP Framework

The proposed WASP is designed to mitigate privacy leakage caused by watchlist attacks in ICN. Unlike conventional approaches that employ static security policies in ICN, the proposed work, WASP, continuously monitors Interest packet traffic. During the monitoring process, it evaluates the privacy exposure associated with each content request. Afterwards, it dynamically activates shielding mechanisms according to the estimated threat level. This adaptive design allows WASP to maintain user privacy and to resist watchlist attacks.

Therefore, the framework of WASP consists of five functional modules: Traffic Monitoring Module, Threat model, Privacy Exposure Estimation Module, Adaptive Shield Controller, Dynamic Cryptographic Protection and Secure Forwarding Engine.

A. Traffic Monitoring Module

The Traffic Monitoring Module continuously observes all incoming Interest packets at every CR. To reduce memory consumption, instead of storing complete traffic histories, WASP maintains a sliding observation window that records only recent communication activities. Let the observation window at time instant t be represented as:

$$\mathcal{W}(t) = \{I_1, I_2, \ldots, I_W\}, \tag{1}$$

where W denotes the sliding window size and I_i represents the ith Interest packet received during the observation period. Equation (1) defines the finite set of Interest packets that are analyzed during every monitoring interval in which privacy analysis will be done. In this regard, for every content object c, the proposed work WASP computes the request frequency as

$$F_c(t) = \sum_{i=1}^{W} \delta(I_i, c), \tag{2}$$

where

$$\delta(I_i, c) = \begin{cases} 1, & \text{if } I_i = c, \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

Equation (2) determines how frequently a content item appears inside the sliding observation window. On the other hand, the indicator function stated in (3) increments the frequency only when the monitored Interest packet corresponds/relevant to the target content. A high value of $F_c(t)$ indicates repeated observations of the same content request, which is one of the primary characteristics of a watchlist attack. In addition, the proposed work, WASP, also considers the timing patterns of content requests. Let the arrival times of a particular content request be represented as

$$t_1, t_2, \ldots, t_n.$$

The average request interval is computed as

$$T_c = \frac{1}{n-1} \sum_{i=2}^{n} (t_i - t_{i-1}), \tag{4}$$

where n denotes the total number of observations. Equation (4) is used to calculate the average time difference between consecutive requests. Meanwhile, a smaller value of T_c indicates that the same content is being requested frequently within the sliding window. Furthermore, to evaluate the cache behavior, WASP computes the cache observation ratio as

$$C_c = \frac{N_{\text{cache}}}{N_{\text{interest}}}, \tag{5}$$

where N_{cache} denotes the number of cache hits and N_{interest} represents the total number of Interest packets associated with content c. Equation (5) measures the proportion of requests satisfied directly by CRs of the network. Excessive cache observations may invoke a watchlist attack by revealing the user's access patterns and leading to increased privacy leakage. Similarly, the average hop count associated with the monitored content is calculated as

$$H_c = \frac{1}{n}\sum_{i=1}^{n} h_i, \tag{6}$$

where h_i represents the number of forwarding hops traversed by the ith Interest packet. Equation (6) characterizes the average routing distance. The parameters considered by the Traffic Monitoring Module collectively provide the necessary information for estimating the privacy exposure of every content request, which is described in the following subsection.

B. Threat Model

The proposed WASP protocol considers a passive adversarial model in which the attacker attempts to infer user access by continuously monitoring Interest packets initiated by the subscriber. Due to the passive nature of the watchlist, this attack produces minimal/negligible network anomalies, making conventional mitigation/detection mechanisms ineffective. The attacker is assumed to compromise one or more FIBs/PITs located along forwarding paths and can record the metadata associated with every traversing Interest packet among CRs.

Let

$$\mathcal{A} = \{a_1, a_2, \ldots, a_m\}, \tag{7}$$

denote the set of observation points controlled by the attacker, where m represents the total number of monitoring locations of CRs present in the network. Equation (7) models the distributed observation capability of the adversary. Each observation point independently collects traffic information for user profiling. The attacker maintains a predefined watchlist containing sensitive content names,

$$\mathcal{L} = \{c_1, c_2, \ldots, c_N\}, \tag{8}$$

where N denotes the number of monitored contents. Whenever an Interest packet requests content item belonging to $\mathcal{L}$, the attacker records the associated communication event together with the corresponding temporal and routing/forwarding information. During every observation interval of sliding window, the attacker collects the sequence

$$\mathcal{O}(t) = \{I_1, I_2, \ldots, I_n\}, \tag{9}$$

where $\mathcal{O}(t)$ denotes all observed Interest packets during time interval t. Using the collected observations from all sliding windows, the attacker estimates the probability that a user has requested a sensitive content object. The corresponding privacy leakage is represented as

$$P_L(c) = \Pr\left(c \mid \mathcal{O}(t)\right), \tag{10}$$

where $P_L(c)$ denotes the inference probability associated with content object c. Equation (10) indicates that the confidence of the attacker increases as more correlated observations become available. The attack objective is therefore formulated as

$$\max \sum_{c\in\mathcal{L}} P_L(c), \tag{11}$$

which maximizes the cumulative privacy information extracted from the monitored watchlist. In contrast, WASP attempts to minimize the observable information available to the attacker while preserving normal network performance. Accordingly, the optimization objective of the proposed protocol is expressed as

$$\min \sum_{c\in\mathcal{L}} P_L(c), \tag{12}$$

subject to

$$PDR \geq PDR_{\min}, \tag{13}$$

$$ISR \geq ISR_{\min}, \tag{14}$$

and

$$L \leq L_{\max}, \tag{15}$$

where L denotes the end-to-end latency. Equations (13)–(15) ensure that privacy enhancement is achieved without significantly degrading communication efficiency. The proposed threat model therefore captures the essential characteristics of practical watchlist attacks, including passive observation, traffic correlation, temporal analysis, and routing inference, thereby providing a realistic basis for evaluating the effectiveness of WASP.

C. Privacy Exposure Estimation

The Traffic Monitoring Module continuously extracts behavioral information of content access from the observed Interest packets. To validate the solidity of the proposed work, some more features are explored to provide not only partial but also full evidence of a watchlist attack. Therefore, WASP combines multiple behavioral characteristics to estimate the overall privacy exposure associated with every content request. In this regard, let the feature vector corresponding to the ith Interest packet be represented as

$$\mathbf{X}_i = [F_i,\ T_i,\ C_i,\ H_i], \tag{16}$$

where

- F_i denotes the request frequency,
- T_i denotes the average inter-arrival time,
- C_i denotes the cache observation ratio, and
- H_i denotes the average hop count.

Equation (16) summarizes the behavioral characteristics extracted from the monitoring sliding window of content access and is helpful in the privacy estimation process. The key problem is having different numerical ranges, such that each feature is normalized using min-max normalization,

$$x_i' = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}}, \tag{17}$$

where x_i denotes the original feature value and x_i' represents the normalized feature. Equation (17) scales every feature into the interval $[0, 1]$. In this way, the normalized features are subsequently combined to compute the Privacy Exposure Score (PES),

$$PES_i = \alpha F_i' + \beta C_i' + \gamma T_i' + \delta H_i', \tag{18}$$

where

$$\alpha, \beta, \gamma, \delta \geq 0,$$

and

$$\alpha + \beta + \gamma + \delta = 1. \tag{19}$$

Equation (18) estimates the cumulative privacy exposure by assigning appropriate importance to each behavioral feature. The weighting constraint in (19) ensures that the score remains properly normalized. A larger value of PES_i indicates that the observed Interest packet exhibits stronger characteristics of watchlist monitoring, whereas smaller values correspond to ordinary communication behavior. Although the weighted score effectively quantifies privacy exposure, it may still vary depending on network traffic conditions. Therefore, the computed score is further normalized as

$$P_i = \frac{PES_i - P_{\min}}{P_{\max} - P_{\min}}, \tag{20}$$

where P_i represents the normalized privacy risk associated with the ith Interest packet. Equation (20) transforms the Privacy Exposure Score into a standardized privacy risk value bounded within the interval $[0, 1]$. This normalization enables the subsequent shielding mechanism to operate consistently under different network loads. Finally, the estimated privacy risk is classified into three security levels according to

$$R_i = \begin{cases} \text{Low}, & P_i < \tau_1, \\ \text{Medium}, & \tau_1 \leq P_i < \tau_2, \\ \text{High}, & P_i \geq \tau_2, \end{cases} \tag{21}$$

where τ_1 and τ_2 denote the lower and upper privacy thresholds, respectively. Equation (21) categorizes the observed communication according to its estimated privacy exposure. Low-risk packets are forwarded normally, whereas medium-and high-risk packets activate the adaptive shielding mechanisms presented in the following subsection. The resulting normalized privacy risk serves as the principal decision variable throughout the WASP framework. It directly controls the shielding intensity, cryptographic protection level, and forwarding strategy, thereby enabling adaptive privacy preservation without introducing unnecessary computational overhead during normal network operation.

D. Adaptive Shield Controller

The privacy risk computed using max-min normalization in Section 4.3 provides a quantitative measure of the likelihood that a subscriber's communication is being monitored by a watchlist adversary. Therefore, being of an adaptive nature, WASP dynamically adjusts the shielding intensity according to the estimated privacy risk. In this way, stronger protection is provided only for high-risk communications. In addition, adaptive shielding intensity can reduce the unnecessary computational overhead of the proposed work. Therefore, to achieve smooth adaptation, the shielding intensity is determined using a sigmoid activation function as stated as,

$$S_i = \frac{1}{1 + \exp\left[-\kappa(P_i - \tau)\right]}, \tag{22}$$

where

- P_i denotes the normalized privacy risk,
- τ represents the shielding threshold,
- κ controls the adaptation rate, and
- $S_i \in [0, 1]$ denotes the shielding intensity.

Equation (22) continuously maps the estimated privacy risk into a shielding intensity. In this way, the protocol categorizes every Interest packet into one of three operational modes,

$$M_i = \begin{cases} \text{Normal}, & S_i < \eta_1, \\ \text{Protected}, & \eta_1 \leq S_i < \eta_2, \\ \text{Shielded}, & S_i \geq \eta_2, \end{cases} \tag{23}$$

where η_1 and η_2 denote the lower and upper shielding thresholds, respectively in Equation (23). To avoid unnecessary activation of cryptographic protection during low-risk conditions, the shielding decision is represented by

$$D_i = \begin{cases} 0, & S_i < \tau, \\ 1, & S_i \geq \tau, \end{cases} \tag{24}$$

where $D_i = 1$ indicates that additional privacy-preserving operations are required. Equation (24) acts as the primary control signal upon the activation of the shielding decision. Afterwards, the Interest packet is forwarded to the Dynamic Cryptographic Protection module; otherwise, the packet follows the standard forwarding procedure. In observation, it is learned that the overall shielding effectiveness depends on the relationship between the estimated privacy risk and the corresponding protection level. This relationship is expressed as follows:

$$E_s = \frac{1}{N} \sum_{i=1}^{N} S_i, \tag{25}$$

where N denotes the total number of monitored Interest packets. Equation (25) represents the average shielding intensity applied across the network by carrying the adaptivity. In the (25), higher values of E_s indicate that the protocol identifies a greater proportion of high-risk communications requiring protection. Furthermore, the protocol regulates the average protection overhead according to

$$O_s = \lambda E_s, \tag{26}$$

where λ represents the computational cost associated with activating additional protection mechanisms. Equation (26) demonstrates that the computational overhead grows proportionally with the average shielding intensity. To deal with this, the proposed work WASP avoids the excessive overhead associated with static always-on security mechanisms while providing stronger protection for high-risk traffic. The Adaptive Shield Controller therefore serves as the central decision-making component of WASP. By continuously transforming the estimated privacy risk into an adaptive protection level, it enables intelligent activation of cryptographic protection and secure forwarding strategies while maintaining high network performance.

E. Dynamic Cryptographic Protection

The Adaptive Shield Controller determines whether additional protection is required for each Interest packet. When the shielding decision is activated, WASP dynamically adjusts the cryptographic strength according to the estimated privacy risk instead of employing a fixed encryption policy. This adaptive strategy provides stronger protection for sensitive communications while minimizing computational overhead during normal network operation. Let the cryptographic protection level associated with the ith Interest packet be denoted by K_i. The protection level is computed as

$$K_i = K_{\min} + (K_{\max} - K_{\min})\, S_i, \tag{27}$$

where

- $K_{\min}$ denotes the minimum cryptographic level,
- $K_{\max}$ denotes the maximum cryptographic level, and
- S_i represents the shielding intensity obtained from (22).

Equation (27) linearly adjusts the cryptographic strength according to the estimated privacy risk. As the shielding intensity increases, stronger cryptographic protection is automatically activated. The encryption operation for the ith Interest packet is represented as

$$C_i = E_{K_i}(M_i), \tag{28}$$

where M_i denotes the original Interest packet and C_i represents the encrypted packet generated using cryptographic key K_i. Equation (28) indicates that the encryption process depends on the dynamically selected protection level rather than a fixed key. At the receiving node, the original packet is recovered through

$$M_i = D_{K_i}(C_i), \tag{29}$$

where $D_{K_i}(\cdot)$ denotes the corresponding decryption operation. Equations (28) and (29) ensure that only legitimate nodes possessing the appropriate cryptographic key can reconstruct the requested content. The computational cost associated with adaptive encryption is modeled as

$$T_{enc} = T_b + \lambda S_i, \tag{30}$$

where T_b denotes the baseline encryption time and λ represents the additional computational cost introduced by adaptive protection. Similarly, the decryption time is expressed as

$$T_{dec} = T_b' + \mu S_i, \tag{31}$$

where T_b' denotes the baseline decryption time and μ represents the corresponding computational coefficient. Equations (30) and (31) indicate that the computational overhead increases gradually with the shielding intensity instead of remaining constant for every communication session. To quantify the randomness introduced by adaptive encryption, the entropy of the encrypted packets is computed as

$$H = -\sum_{j=1}^{m} p_j \log_2 p_j, \tag{32}$$

where p_j denotes the probability of occurrence of the jth encrypted symbol. Equation (32) measures the uncertainty associated with the encrypted traffic. Larger entropy values indicate stronger resistance against statistical inference and traffic analysis performed by watchlist adversaries. Finally, the overall cryptographic overhead introduced by WASP is defined as

$$O_c = T_{enc} + T_{dec}, \tag{33}$$

where O_c represents the total processing overhead incurred by adaptive cryptographic protection. Equation (33) demonstrates that the overall computational cost depends on the dynamically selected protection level rather than the total network traffic. Consequently, WASP significantly reduces unnecessary encryption overhead while maintaining effective privacy preservation against watchlist attacks. The dynamically protected Interest packets are subsequently transmitted to the Secure Forwarding Engine, which determines the forwarding strategy according to the estimated privacy exposure.

F. Secure Forwarding Strategy

Following the Dynamic Cryptographic Protection stage, the protected Interest packets are processed by the Secure Forwarding Engine. The primary objective of WASP is to reduce the information available to a corresponding attacker. Let the forwarding probability associated with the ith Interest packet be denoted by $P_f(i)$. The forwarding probability is defined as

$$P_f(i) = 1 - S_i, \tag{34}$$

where S_i is the shielding intensity obtained from (22). Equation (34) establishes an inverse relationship between privacy risk and forwarding probability. From this relationship, it can be said that Interest packets exhibiting low privacy risk are forwarded using the normal routing strategy. On the other hand, highly sensitive requests activate additional forwarding protection. The forwarding decision is determined according to

$$F_i = \begin{cases} \text{Normal Forwarding}, & D_i = 0, \\ \text{Secure Forwarding}, & D_i = 1, \end{cases} \tag{35}$$

where D_i is the shielding decision computed in (24). Equation (35) separates communication without WASP from privacy-sensitive communication. In this regard, it can be seen that only packets classified as high-risk activate the secure forwarding mechanism. To reduce traffic correlation, WASP introduces an adaptive forwarding delay given by

$$\Delta_i = \Delta_{\min} + (\Delta_{\max} - \Delta_{\min}) S_i, \tag{36}$$

where $\Delta_{\min}$ and $\Delta_{\max}$ denote the minimum and maximum forwarding delays, respectively. Equation (36) introduces balanced randomness into packet transmission. The reason is that forwarding/routing delay depends on the estimated privacy risk and repetitive traffic patterns. In this way, it is significantly more difficult for a watchlist adversary to correlate among caching patterns. Similarly, the routing diversity factor is computed as

$$R_i = 1 + \rho S_i, \tag{37}$$

where ρ is the routing adaptation coefficient. Equation (37) increases routing diversity as the privacy risk increases at each CR of ICN. Consequently, repeated Interest packets from multiple subscribers requesting the same content are less likely to traverse identical network paths. Therefore, reducing the effectiveness of long-term traffic observation. Motivated by this, the forwarding cost associated with the selected route is expressed as

$$C_f = \omega_1 L + \omega_2 H + \omega_3 (1 - PDR), \tag{38}$$

where

- L denotes the end-to-end latency,
- H represents the average hop count,
- PDR denotes the packet delivery ratio, and
- $\omega_1 + \omega_2 + \omega_3 = 1$.

Equation (38) evaluates the quality of a forwarding path by jointly considering communication delay, routing efficiency, and packet delivery performance.

$$R^* = \arg\min C_f, \tag{39}$$

where R^* denotes the forwarding route that minimizes the communication cost along with the proposed working strategy as stated in Equation (39). Finally, the overall forwarding efficiency of the proposed protocol is defined as

$$E_f = \frac{ISR \times PDR}{L \times H}, \tag{40}$$

where ISR, PDR, L, and H represent the Interest Satisfaction Rate, Packet Delivery Ratio, end-to-end latency, and average hop count, respectively. Equation (40) jointly evaluates the effectiveness of the forwarding strategy by considering satisfaction score, communication delay, speed and routing efficiency. Larger values of E_f indicate better forwarding performance in addition to privacy protection. The proposed Secure Forwarding Strategy therefore complements the adaptive shielding and cryptographic protection mechanisms by dynamically selecting forwarding policies that minimize traffic correlation without significantly affecting normal network operation.

G. WASP Algorithm

Algorithm 1 summarizes the complete operational procedure of the proposed work WASP. The protocol continuously monitors incoming Interest packets generated by the different subscribers. Afterwards, estimation of the associated privacy exposure and computation of the adaptive shielding intensity have been made. These measures dynamically select the cryptographic protection level and finally perform secure forwarding according to the estimated privacy risk.

Algorithm 1 Watchlist Attack Shield Protocol (WASP)

Require: Incoming Interest packet I_i
Ensure: Secure forwarding decision
Receive Interest packet I_i
Update sliding observation window $\mathcal{W}(t)$
Compute request frequency using (2)
Compute average request interval using (4)
Compute cache observation ratio using (5)
Compute average hop count using (6)
Construct feature vector $\mathbf{X}_i$ using (16)
Normalize all traffic features using (17)
Compute Privacy Exposure Score using (18)
Compute normalized privacy risk using (20)
Determine privacy risk level using (21)
Compute shielding intensity using (22)
Determine operational mode using (23)
Compute shielding decision using (24)
if $D_i = 1$ **then**
    Select adaptive cryptographic level using (27)
    Encrypt Interest packet using (28)
    Estimate encryption overhead using (30)
    Estimate entropy using (32)
    Compute forwarding probability using (34)
    Compute adaptive forwarding delay using (36)
    Evaluate forwarding cost using (38)
    Select optimal forwarding path using (39)
    Securely forward encrypted Interest packet
else
    Forward Interest packet using standard ICN forwarding
end if
Receive corresponding Data packet
if $D_i = 1$ **then**
    Decrypt Data packet using (29)
end if
Update monitoring statistics
Return Data packet to consumer

H. Computational Complexity and Security Analysis

This subsection analyzes the computational complexity of the proposed WASP protocol and discusses its security properties against watchlist attacks. Since WASP performs adaptive privacy protection only for high-risk communications, the computational overhead remains significantly lower than that of static security mechanisms.

1) Computational Complexity Analysis

As mentioned in previously discussed subsections, the overall computational cost of WASP is determined by five major processing stages: traffic monitoring, privacy exposure estimation, adaptive shielding, dynamic cryptographic protection, and secure forwarding. To calculate the cost, Traffic Monitoring Module scans all Interest packets contained in the sliding observation window. In this regrad, let the observation window contain W Interest packets. The computational complexity of feature extraction can be calculated as:

$$C_{TM} = O(W). \tag{41}$$

Equation (41) indicates that traffic monitoring grows linearly with the observation sliding window size. Assume that the second step named Privacy Exposure Estimation Module, utilizes f behavioral features. Since each feature is normalized using max-min and processed exactly once, the computational complexity can be defined as:

$$C_{PE} = O(f). \tag{42}$$

As the number of extracted features is fixed in the proposed framework, Equation (42) represents constant computational overhead during practical deployment. The Adaptive Shield Controller evaluates the sigmoid function presented in Equation (22) for each Interest packet. Since the computation involves only elementary arithmetic operations, its complexity is

$$C_{AS} = O(1). \tag{43}$$

Similarly, the Dynamic Cryptographic Protection Module performs adaptive key selection and packet encryption. Let k denote the cryptographic key length. The corresponding complexity is expressed as

$$C_{CP} = O(k). \tag{44}$$

Equation (44) indicates that cryptographic processing increases proportionally with the selected security level. The Secure Forwarding Engine selects the forwarding path from the available routing entries. Assuming that routing information is maintained using efficient lookup structures, the forwarding complexity becomes

$$C_{SF} = O(\log N), \tag{45}$$

where N denotes the number of forwarding entries. Combining all processing stages, the overall computational complexity of WASP is

$$C_{\text{WASP}} = O(W + f + k + \log N). \tag{46}$$

Equation (46) demonstrates that the proposed protocol exhibits polynomial-time complexity and therefore scales efficiently for large network deployments.

2) Security Analysis

The primary objective of WASP is to reduce/eliminate/hide the information available to passive watchlist attackers while maintaining low communication overhead with high PDR. Since the attacker attempts to maximize the privacy leakage defined in Equation (11), the proposed protocol WASP minimizes the same objective through adaptive shielding and secure forwarding. Consequently,

$$P_L^{\text{WASP}} < P_L^{\text{Attack}}, \tag{47}$$

where P_L^{Attack} and P_L^{WASP} denote the privacy leakage without and with the proposed protocol as denoted in Equation (47), respectively. This mathematical foundation shows that the attack is significantly reduced after applying adaptive shielding. Furthermore, the probability of successful user profiling decreases as the shielding intensity increases. This relationship is expressed as

$$P_{\text{profile}} = P_L(c)(1 - S_i), \tag{48}$$

where S_i is the shielding intensity computed in Equation (22) and used in Equation (48) to define the coorelation. Afterwards, an entrophy is calculated to estimate the uncertainty associated with encrypted traffic used in Equation (32). Larger entropy values reduce the statistical information available for traffic analysis and consequently improve resistance against watchlist attacks. Finally, WASP preserves communication performance by satisfying the optimization constraints defined in Equations (13)–(15). Therefore, the proposed protocol simultaneously minimizes privacy leakage while maintaining various key performance metrices. In addition, theoretical analysis further confirms that WASP provides adaptive privacy preservation with low computational complexity while effectively mitigating passive watchlist attacks.

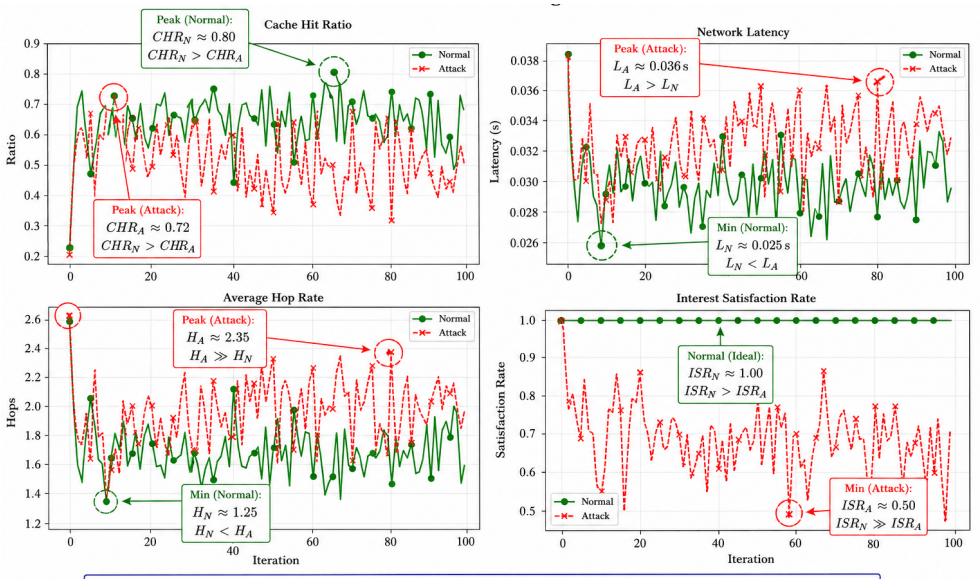


Fig. 1. Impact of the Watchlist attack on CHR, latency, hop count, and ISR under normal, attack, and WASP-protected conditions.

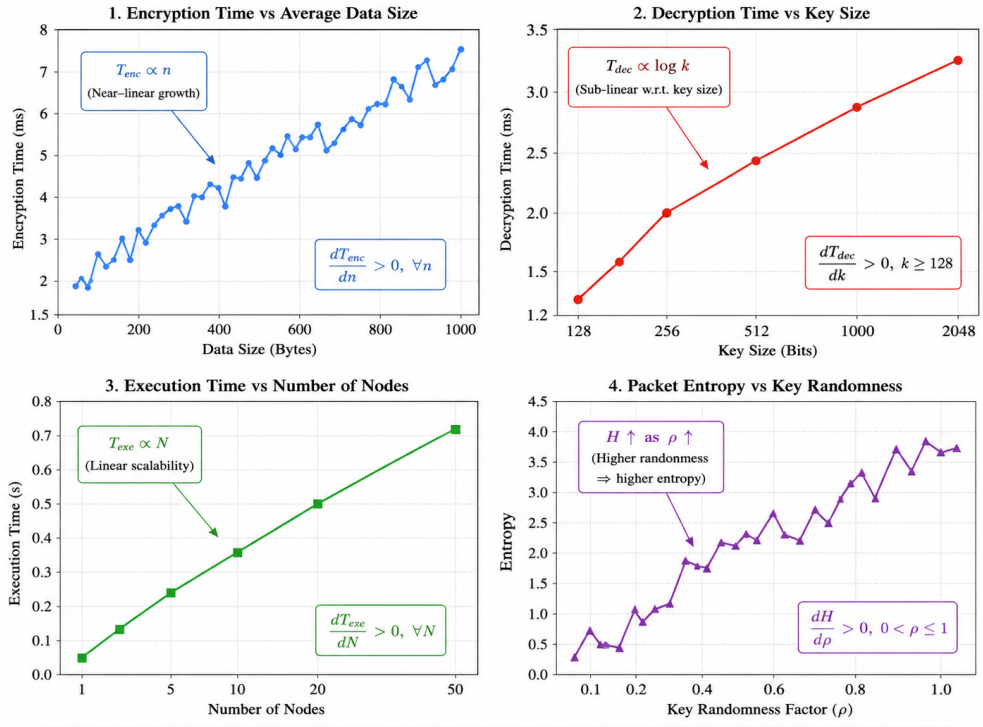


Fig. 2. Performance analysis of the proposed adaptive cryptographic framework. (a) Encryption time increases approximately linearly with data size ($T_{\text{enc}} \propto n$). (b) Decryption time exhibits sub-linear growth with increasing key size ($T_{\text{dec}} \propto \log k$). (c) Execution time scales nearly linearly with the number of participating nodes ($T_{\text{exe}} \propto N$). (d) Packet entropy increases with key randomness ($\frac{dH}{d\rho} > 0$), indicating improved cryptographic diffusion and stronger randomness.

IV. Experimental Setup

The proposed WASP protocol was implemented and evaluated using the PYcarus simulator running on NS-3.35. The simulated ICN topology comprises subscribers, CRs, and producers/servers, in which each CR maintains PIT, CS, and FIB for content forwarding and in-network caching. Consumers generated Interest packets according to a Zipf distribution ($\alpha = 0.8$) to emulate realistic content popularity. The observation window size for traffic monitoring was fixed at 50 Interest packets, and adaptive forwarding was employed throughout the network.

Fig. 1 demonstrates the effect of the Watchlist Attack on four key network performance metrics, with normal operation and attack scenario over 100 iterations [17]. The plot of the CHR indicates that the normal network always has a higher score (with a maximum value of around $\text{CHR}_N \approx 0.80$). On the other hand, a relevant attack reduces the cache efficiency to about $\text{CHR}_A \approx 0.72$, satisfying $\text{CHR}_N > \text{CHR}_A$. The reduction of nearly 0.08 (approximately 11.1%) indicates that malicious monitoring and repeated requests disturb cache locality. It has outcomes that reduce the probability of retrieving content directly from nearby CR caches. The network latency analysis further demonstrates that the watchlist attack significantly increases communication delay, where the minimum latency under normal operation is approximately $L_N \approx 0.025$ s, while the attack increases the latency to nearly $L_A \approx 0.036$ s, satisfying $L_A > L_N$. A similar degradation is observed in the routing efficiency shown by the average hop count. Under normal conditions, packets

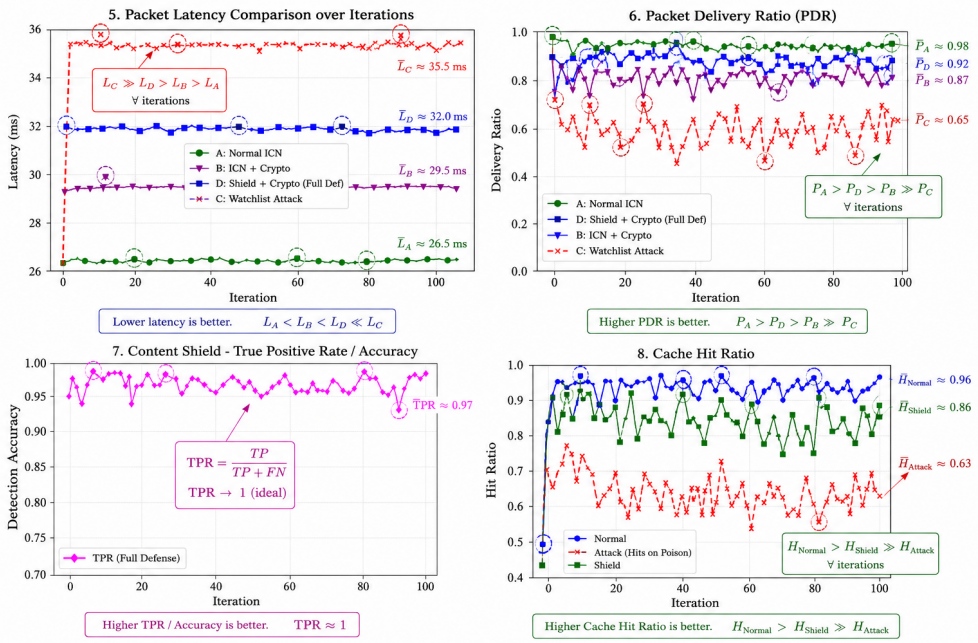


Fig. 3. Performance comparison under different network conditions. (a) Packet latency comparison over iterations, where $L_A < L_B < L_D \ll L_C$ indicates that the proposed full-defense framework maintains significantly lower latency than the watchlist attack. (b) PDR, showing $P_A > P_D > P_B \gg P_C$ throughout the experiment. (c) Detection accuracy of the Content Shield, where $\text{TPR} = \frac{TP}{TP+FN} \approx 0.97$ demonstrates consistently high detection capability. (d) CHR, satisfying $H_{\text{Normal}} > H_{\text{Shield}} \gg H_{\text{Attack}}$, indicating that the defense preserves cache efficiency while substantially outperforming the attack scenario. Circled markers highlight representative extrema and stable operating points.

traverse the network using approximately $H_N \approx 1.25$ hops on average, whereas the attack increases the routing distance to approximately $H_A \approx 2.35$ hops, satisfying $H_A > H_N$. The results shown highlight representative peak and minimum operating points, collectively confirming that the Watchlist attack simultaneously decreases cache efficiency and content availability while increasing routing overhead and communication latency. Consequently, the observed relationships $\text{CHR}_N > \text{CHR}_A$, $L_N < L_A$, $H_N < H_A$, and $\text{ISR}_N \gg \text{ISR}_A$ quantitatively demonstrate the severe degradation of network performance caused by continuous traffic observation and user-interest inference.

V. Performance Evaluation

This section evaluates the performance of the proposed WASP protocol under watchlist attack scenarios. The plotted results are compared with existing state-of-the-art security mechanisms [17], [18] as discussed in the literature survey to analyze the performance in terms of various key indicators. The performance of proposed work WASP is denoted by Shield+Crypto in plots.

Fig. 2 presents the computational and cryptographic performance of the proposed framework under varying operational conditions. It can be observed from Fig. 2(a) that the encryption time exhibits an approximately linear relationship with the data size. In this regard, the mathematical formulation $T_{\text{enc}} \propto n$, indicates that the computational overhead increases proportionally with the amount of data being protected. This behavior is consistent with the positive gradient $\frac{dT_{\text{enc}}}{dn} > 0$. In this way, it confirms that the proposed mechanism scales efficiently without exhibiting exponential computational growth. Fig. 2(b) illustrates the corresponding decryption performance for different key sizes. It can be stated that as the size of the cryptographic algorithm increases from 128 bits to 2048 bits, the decryption time is observed to rise from approximately 1.25 ms to 3.25 ms. Mathematically, this increase has a logarithmic trend $T_{\text{dec}} \propto \log(k)$. The monotonic increase $\frac{dT_{\text{dec}}}{dk} > 0$ implies a predictable scalability even for high-security configurations. Moreover, Fig. 2(c) evaluates the scalability of the proposed WASP with respect to the number of participating nodes in ICN. The execution time increases almost linearly from approximately 0.05 s for a single node to about 0.72 s for 50 nodes, satisfying the relationship $T_{\text{exe}} \propto N$. Finally, Fig. 2(d) analyzes the entropy of encrypted packets as a function of the key randomness factor ρ. As ρ increases from 0.05 to 1.0, the packet entropy rises from approximately 0.3 to nearly 3.8, satisfying $\frac{dH}{d\rho} > 0$. The increase in entropy indicates improved diffusion and greater statistical uncertainty within the encrypted packets, thereby making traffic analysis and watchlist-based inference substantially more difficult.

Fig. 3 demonstrates the effectiveness of the proposed WASP framework under four representative network conditions. Fig. 3(a) is used to represent the average packet latency which remains nearly constant throughout the 100 iterations. In compared with the normal network, which achieves the lowest delay of $\overline{L}_A \approx 26.5$ ms, as compared to the proposed WASP framework ($\overline{L}_D \approx 32.0$ ms), and the watchlist attack exhibiting the highest latency of $\overline{L}_C \approx 35.5$ ms. Consequently, the latency ordering satisfies $L_A < L_B < L_D \ll L_C$, indicating that although adaptive shielding introduces approximately 5.5 ms of additional processing compared with normal forwarding. Fig. 3(b) evaluates the PDR, where the normal network consistently maintains $\overline{P}_A \approx 0.98$, while the proposed full-defense mechanism preserves a high delivery ratio of $\overline{P}_D \approx 0.92$, outperforming ICN with conventional cryptography ($\overline{P}_B \approx 0.87$) and significantly exceeding the attack condition ($\overline{P}_C \approx 0.65$). This ordering satisfies $P_A > P_D > P_B \gg P_C$, demonstrating that adaptive shielding successfully activates the privacy-preserving operations. Fig. 3(c) is used to illustrate the detection capability of the proposed WASP using Content Shield feature, where the True Positive Rate remains highly stable around $\text{TPR} \approx 0.97$, using the formula $\text{TPR} = \frac{TP}{TP+FN}$. Furthermore, Fig. 3(d) compares the CHR metric of ICN under different operating conditions. It can be seen that the normal network achieves an average CHR of approximately $\overline{H}_{\text{Normal}} \approx 0.96$. On the other hand, WASP maintains $\overline{H}_{\text{Shield}} \approx 0.86$, substantially higher than the attack scenario with $\overline{H}_{\text{Attack}} \approx 0.63$. Therefore, the relationship denoted by the mathmatical formula $H_{\text{Normal}} > H_{\text{Shield}} \gg H_{\text{Attack}}$ defines that WASP preserves nearly 90% of the cache efficiency of normal operation while improving the CHR by approximately 36.5% over the watchlist attack $\left(\frac{0.86-0.63}{0.63} \times 100\%\right)$.

Fig. 4 compares the most important metric ISR, and average

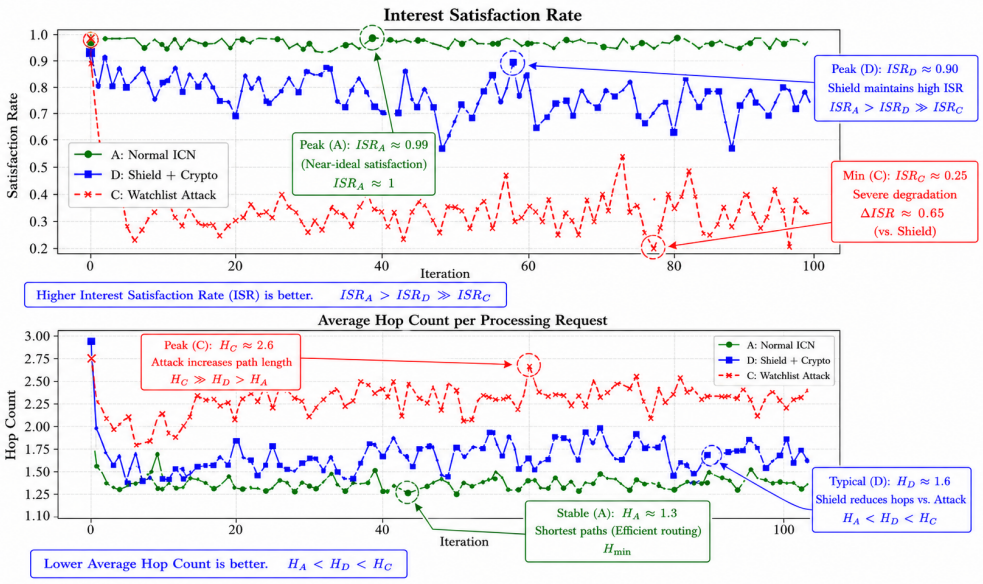


Fig. 4. Performance comparison of the proposed Content Shield under a watchlist attack. (a) ISR, where $\mathrm{ISR}_A > \mathrm{ISR}_D \gg \mathrm{ISR}_C$, demonstrating that the proposed Shield+Crypto framework preserves a high content retrieval success rate compared with the attack scenario. (b) Average Hop Count, satisfying $H_A < H_D \ll H_C$, indicating that the proposed defense maintains efficient forwarding paths while significantly reducing the routing overhead introduced by the watchlist attack. Circled points highlight representative extrema and stable operating regions.

hop count. As illustrated in Fig. 4(a), it can be observed that the normal network maintains $\mathrm{ISR}_A \approx 0.97$–$0.99$. However, while the proposed Shield+Crypto framework utilized by WASP consistently preserves a high satisfaction rate of approximately $\mathrm{ISR}_D \approx 0.75$–$0.90$. In contrast, the watchlist attack without a detection/mitigation strategy severely degrades network performance, reducing the ISR to approximately $\mathrm{ISR}_C \approx 0.20$–$0.55$, with the minimum observed value close to 0.25. Consequently, the relationship $\mathrm{ISR}_A > \mathrm{ISR}_D \gg \mathrm{ISR}_C$ holds throughout the experiment, indicating that the proposed defense maintains nearly 80% of the normal retrieval performance while improving the satisfaction rate by approximately $\Delta\mathrm{ISR} \approx 0.65$ over the attack scenario. Fig. 4(b) further evaluates routing efficiency using the average hop count. The normal network consistently achieves the shortest forwarding paths with $H_A \approx 1.30$, whereas the proposed Shield+Crypto mechanism introduces only a modest routing overhead with $H_D \approx 1.60$, corresponding to an increase of approximately 0.30 hops ($\approx 23\%$) compared with normal operation. Under the watchlist attack, however, the average hop count increases significantly to $H_C \approx 2.20$–2.60, satisfying the relationship $H_A < H_D \ll H_C$. Compared with the attack scenario, the proposed framework reduces the forwarding path length by approximately one hop ($\Delta H \approx 0.9$), corresponding to nearly a 40% reduction in routing overhead $\left(\frac{2.6-1.6}{2.6} \times 100\%\right)$. The circled points highlight representative extrema and stable operating regions, demonstrating that although the proposed defense introduces a small forwarding overhead to preserve user privacy, it maintains high content retrieval efficiency while substantially reducing the routing inefficiency caused by watchlist attacks. Collectively, the results verify that the proposed WASP utlized Shield+Crypto framework effectively balances privacy preservation and network performance by maintaining a high ISR together with short forwarding paths even under continuous adversarial monitoring.

VI. Conclusion and Future Scope

In this paper, an adaptive privacy-preserving framework WASP is proposed to mitigate watchlist attacks in ICN. Unlike conventional security mechanisms that employ static cryptographic protection, the proposed work WASP integrates privacy exposure estimation, adaptive shielding, dynamic cryptographic protection, and secure forwarding. The proposed protocol continuously monitors network behavior, estimates the privacy risk associated with each Interest packet generated by the subscribers, and selectively activates protection mechanisms according to the observed threat level. This adaptive activation of the security mechanism significantly reduces unnecessary computational overhead while preserving the user's privacy. Comprehensive mathematical modeling was developed to characterize the threat model to provide a rigorous analytical foundation for the proposed protocol WASP. Extensive simulations were conducted and validated using the topological parameters that demonstrate the effectiveness of WASP under watchlist attack scenarios. Experimental results show that the proposed framework maintains high packet delivery performance ($\mathrm{PDR} \approx 0.92$), consistently achieves a True Positive Rate of approximately 0.97, preserves a CHR of nearly 0.86, and maintains an ISR above 0.80 while introducing only a modest increase in communication latency compared with normal network operation. Future work will focus on extending WASP to distributed and heterogeneous edge environments where multiple routers collaboratively exchange privacy intelligence while preserving scalability. The framework will be enhanced using lightweight federated learning and graph neural networks to improve the identification of coordinated and distributed watchlist attacks without exposing sensitive traffic information.

References

[1] Bukhowah, Rawan, Ahmed Aljughaiman, and MM Hafizur Rahman. "Detection of dos attacks for IoT in information-centric networks using machine learning: Opportunities, challenges, and future research directions." Electronics 13, no. 6 (2024): 1031.

[2] Ari, Ado Adamou Abba, Youssoufa Hara Soukolsou, Sammy Salim Daissinta Baidi, Moussa Aboubakar, Nabila Labraoui, Alidou Mohamadou, and Ousmane Thiare. "A Thorough Review of Security in Information-Centric Networking." IET Information Security 2025, no. 1 (2025): 5178335.

[3] Bilal, Muhammad, and Sangheon Pack. "Secure distribution of protected content in information-centric networking." IEEE Systems Journal 14, no. 2 (2019): 1921-1932.

[4] Mannes, Elisa, and Carlos Maziero. "Naming content on the network layer: A security analysis of the information-centric network model." ACM Computing Surveys (CSUR) 52, no. 3 (2019): 1-28.

[5] Fang, Weidong, Ming Xu, Chunsheng Zhu, Weili Han, Wuxiong Zhang, and Joel JPC Rodrigues. "FETMS: Fast and efficient trust management scheme for information-centric networking in Internet of Things." IEEE access 7 (2019): 13476-13485.

[6] Nour, Boubakr, Spyridon Mastorakis, Rehmat Ullah, and Nicholas Stergiou. "Information-centric networking in wireless environments: Security risks and challenges." IEEE Wireless Communications 28, no. 2 (2021): 121-127.

[7] Tian, Yi, Dapeng Man, Hanbo Wang, Huanran Wang, and Wu Yang. "Hybrid-Based Timing Attack for Path Inference in Named Data Networking." In 2024 20th International Conference on Mobility, Sensing and Networking (MSN), pp. 529-537. IEEE Computer Society, 2024.

[8] G. Xylomenos *et al.*, "A survey of information-centric networking research," *IEEE Communications Surveys & Tutorials*, vol. 16, no. 2, pp. 1024–1049, 2014.

[9] B. Ahlgren, C. Dannewitz, C. Imbrenda, D. Kutscher, and B. Ohlman, "A survey of information-centric networking," *IEEE Communications Magazine*, vol. 50, no. 7, pp. 26–36, Jul. 2012.

[10] K. Yu *et al.*, "Information-centric networking: Research and standardization status," *IEEE Access*, vol. 7, pp. 126164–126176, 2019.

[11] E. G. AbdAllah, H. S. Hassanein, and M. Zulkernine, "A survey of security attacks in information-centric networking," *IEEE Communications Surveys & Tutorials*, vol. 17, no. 3, pp. 1441–1454, 2015.

[12] R. Tourani, T. Mick, S. Misra, and G. Panwar, "Security, privacy, and access control in information-centric networking: A survey," *IEEE Communications Surveys & Tutorials*, vol. 20, no. 1, pp. 566–600, 2018.

[13] M. Zhang, H. Luo, and H. Zhang, "A survey of caching mechanisms in information-centric networking," *IEEE Communications Surveys & Tutorials*, vol. 17, no. 3, pp. 1473–1499, 2015.

[14] A. Mohaisen, H. Mekky, X. Zhang, H. Xie, and Y. Kim, "Timing attacks on access privacy in information centric networks and countermeasures," *IEEE Transactions on Dependable and Secure Computing*, vol. 12, no. 6, pp. 675–687, 2015.

[15] G. Acs, M. Conti, P. Gasti, C. Ghali, and G. Tsudik, "Privacy-aware caching in information-centric networking," *IEEE Transactions on Dependable and Secure Computing*, vol. 16, no. 2, pp. 313–328, 2019.

[16] M. Anisetti *et al.*, "A security certification scheme for information-centric networks," *IEEE Transactions on Network and Service Management*, vol. 19, no. 3, pp. 2397–2412, 2022.

[17] Sivaraman, Vignesh, and Biplab Sikdar. "A defense mechanism against timing attacks on user privacy in ICN." IEEE/ACM Transactions on Networking 29, no. 6 (2021): 2709-2722.

[18] Bardhi, Enkeleda, Mauro Conti, Riccardo Lazzeretti, and Eleonora Losiouk. "ICN PATTA: ICN privacy attack through traffic analysis." In 2021 IEEE 46th Conference on Local Computer Networks (LCN), pp. 443-446. IEEE, 2021.

[19] Mirajkar, Riddhi, Gitanjali Shinde, Parikshit Mahalle, and Nilesh Sable. "Threat Analysis and Attack Modeling in Data-Centric Communication for Named Data Networking." In Demystifying AI and ML for Cyber–Threat Intelligence, pp. 483-498. Cham: Springer Nature Switzerland, 2025.